Problem 1

The empirical coverage on the test set is about 0.897, and the average size of the conformal sets is 1.732. The marginal coverage guarantee means that, over many test samples, the probability that the true label is contained within the prediction set is at least 1-alpha.

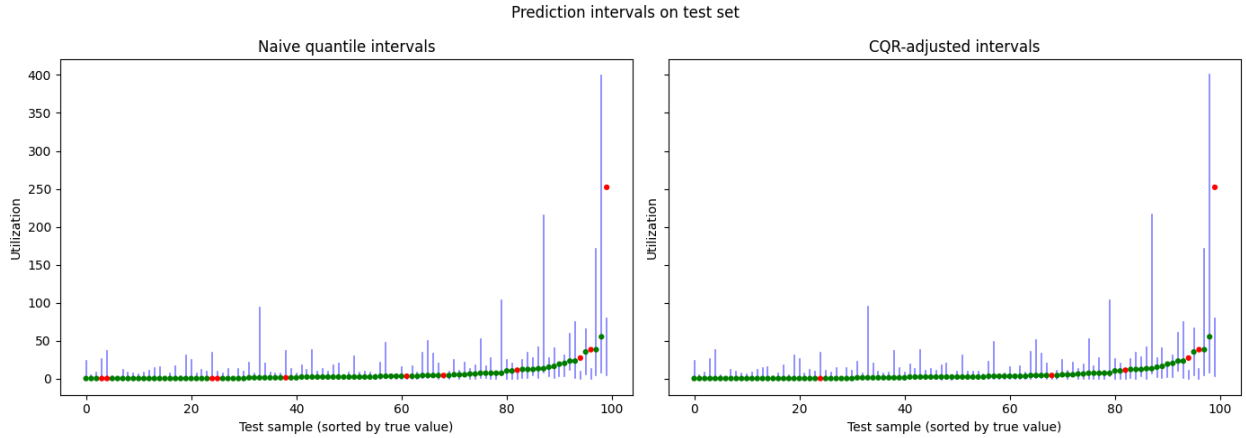
For the left image, the prediction set is: Ocarina, and for the right image, the prediction set is Cardigan Welsh Corgi.

Problem 2

Q1

The empirical coverage for the naive quantile intervals is about 0.853, which is below the target coverage level of 0.9.

The empirical coverage for the CQR intervals is about 0.906, which has reached the target coverage rate.



Above is the visualization of the naive quantile intervals and CQR-adjusted intervals, along with some points randomly chosen from the samples. The green points mean that the sample is

covered by the conformal set, and the red points mean that the sample is not covered. From the graph we can see that although the width of the intervals do not change significantly, there are fewer red points for the CQR-adjusted coverage compared to the naive coverage, which indicates an improvement in the coverage rate.

Q2

	Naive coverage	CQR coverage
Group 0 (0-18)	0.919	0.950
Group 1 (18-65)	0.835	0.900
Group 2 (65-85)	0.782	0.830

The empirical coverage for the three age groups shows clear differences across groups. Overall, CQR coverage performs better on every group, but the group disparity still exists.

These results illustrate the difference between marginal and conditional coverage. Conformal methods guarantee marginal coverage, meaning that the overall coverage across all samples is at least the target level. However, this guarantee does not necessarily hold for specific subgroups. In particular, the oldest group (65–85) has much lower coverage compared to the youngest group.

Unequal coverage across demographic groups raises fairness concerns, especially in clinical settings. For example, if older patients consistently receive lower coverage, the prediction intervals may be less reliable for this group, potentially leading to worse decision-making.

Q3

	Conformal quantile	CQR coverage	Interval width
Group 0 (0-18)	-0.101	0.900	13.797
Group 1 (18-65)	0.656	0.876	23.000
Group 2 (65-85)	5.821	0.934	72.099

The aggregated empirical coverage is about 0.903. The empirical coverage by age group is shown above.

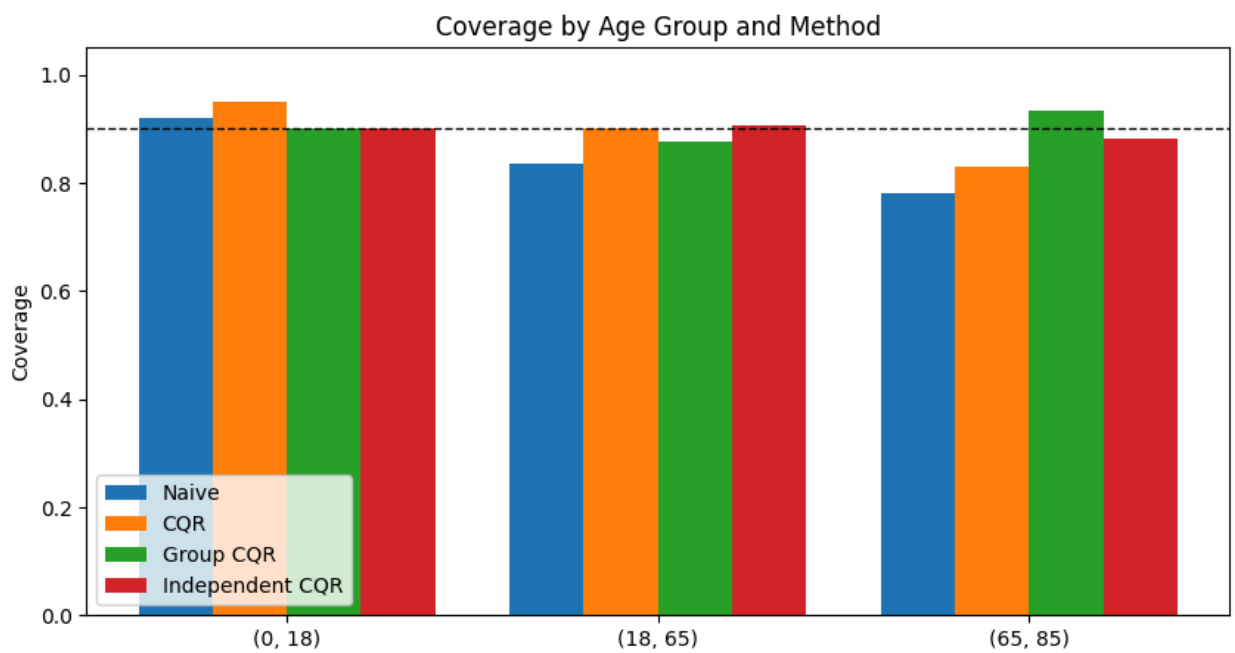
Groups with larger conformal quantiles tend to have wider prediction intervals. This is because a larger quantile means the model tends to make greater errors, which expands the interval more.

Also, after the fix, the coverage of the oldest group (65-85) becomes the highest. Maybe this is achieved by significantly widening the interval. As shown in the table, the interval width for group 2 is about 72, a large increase compared to other groups. By returning a large interval, the model becomes more conservative so that it can achieve the target coverage rate. However, it does not necessarily mean the model performs well in this group.

Q4

	Conformal quantile	CQR coverage	Interval width
Group 0 (0-18)	0.927	0.900	11.328
Group 1 (18-65)	0.894	0.906	20.480
Group 2 (65-85)	3.317	0.883	74.610

The aggregated coverage is approximately 0.896, slightly lower than our target. The empirical coverage by age group, each group's quantile, and their interval width is shown in the above table. The oldest group still has slightly lower coverage compared to the others.



Compared to the groupwise CQR method in Q3, the independent model method does not consistently improve the coverage rate. The overall coverage is also lower.

This suggests that training separate models for each group does not necessarily lead to better performance. One possible reason is that splitting the data reduces the amount of training and

calibration data available for each group, especially for smaller groups, leading to less stable estimates of both the model and the conformal quantile.

Also, the interval width for the oldest group remains significantly larger (74.610), indicating higher uncertainty in predictions for this group. This pattern is consistent with previous methods and suggests that the difficulty of prediction varies across age groups.

Q5

Fix 1 has a smaller interval for the oldest group, which is the hardest to predict, while Fix 2 has smaller intervals for group 0 and 1. Fix 1 is generally more efficient when data is limited, because it uses a single model trained on all data, leading to more stable predictions.

I would recommend Fix 1. It provides more reliable predictions for elderly people (the group that tends to use more healthcare resources) while making better use of all available data. This is important in healthcare settings, where overly wide intervals may reduce the usefulness of predictions.

Problem 3

(a)

Q1

Age distribution:

	median	mean	min	max
A	51	50.173	0	100
B	33	35.735	18	76
C	36	41.579	18	96
D	39	44.153	18	96
Ground truth	32	35.143	18	96

Method A (random mode) performs the worst. Its statistics are far from the ground truth. This is expected because random mode does not preserve the underlying data distribution.

Method B (independent attributes, $\varepsilon=0.1$) is much closer to the ground truth. Both mean and median are very close, indicating that modeling each attribute independently already captures basic marginal distributions well.

Methods C and D (correlated attributes) show higher mean and median values than the ground truth. Although they attempt to model dependencies between attributes, they introduce some bias in the distribution of age. Between them, method D (k=2) is slightly closer than method C (k=1), suggesting that modeling more dependencies can improve accuracy.

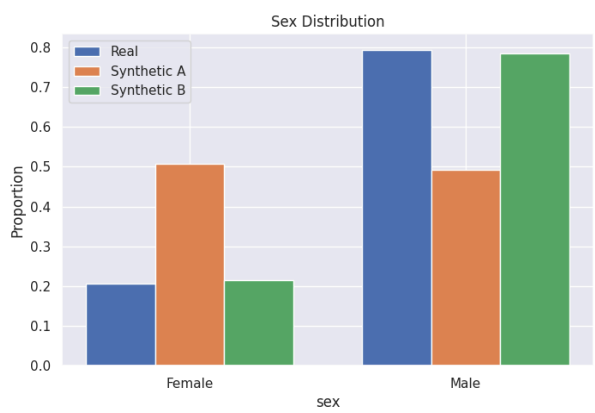
Score distribution:

	median	mean	min	max
A	5	4.939	-1	10
B	4	4.366	1	10
C	5	4.949	-1	10
D	4	4.466	-1	10
Ground truth	4	4.371	-1	10

For the score attribute, methods B, C, and D all perform relatively well. Their means and medians are close to the ground truth, indicating that the distribution of score is easier to capture.

Method A again performs worse due to randomness, but the deviation is smaller compared to age.

Q2



The distribution generated by random mode (A) shows a relatively uniform distribution in both age and sex. However, in the real dataset, there are significantly less females than males, and the age distribution is skewed to the right. The independent attribute mode (B) generates a distribution that is very similar to the real distribution.

KS test (age):

A: 0.3735 B: 0.0263

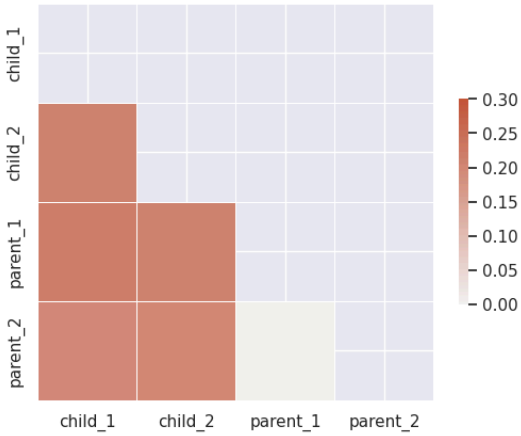
KL divergence (sex):

A: 0.2232 B: 0.0002

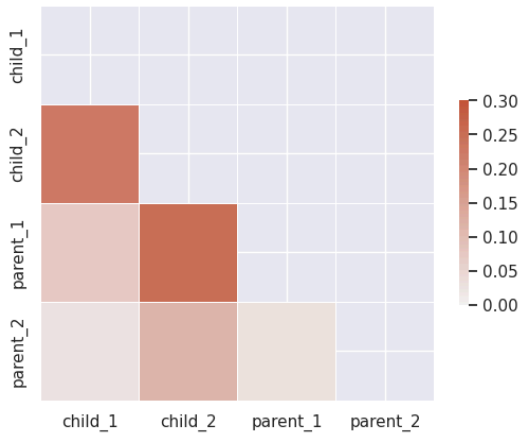
For the age attribute, the KS statistic of method A is much larger than that of method B, indicating that the age distribution generated by A deviates significantly from the real distribution. In contrast, method B produces a distribution that is very close to the real data.

For the sex attribute, the KL divergence of method A is relatively large, while method B achieves a value very close to zero. This shows that method B accurately captures the categorical distribution of sex, whereas method A fails to do so.

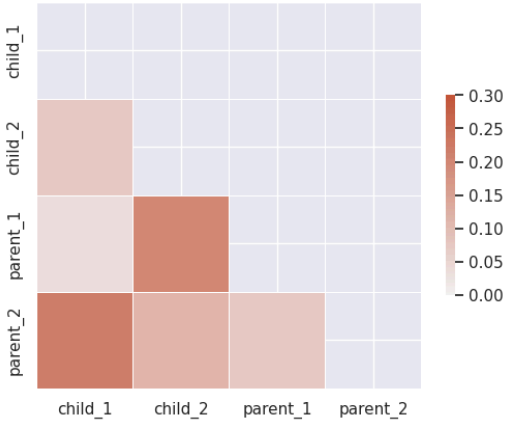
Q3



Mutual Information for df_fake



Mutual Information for df_fake_C



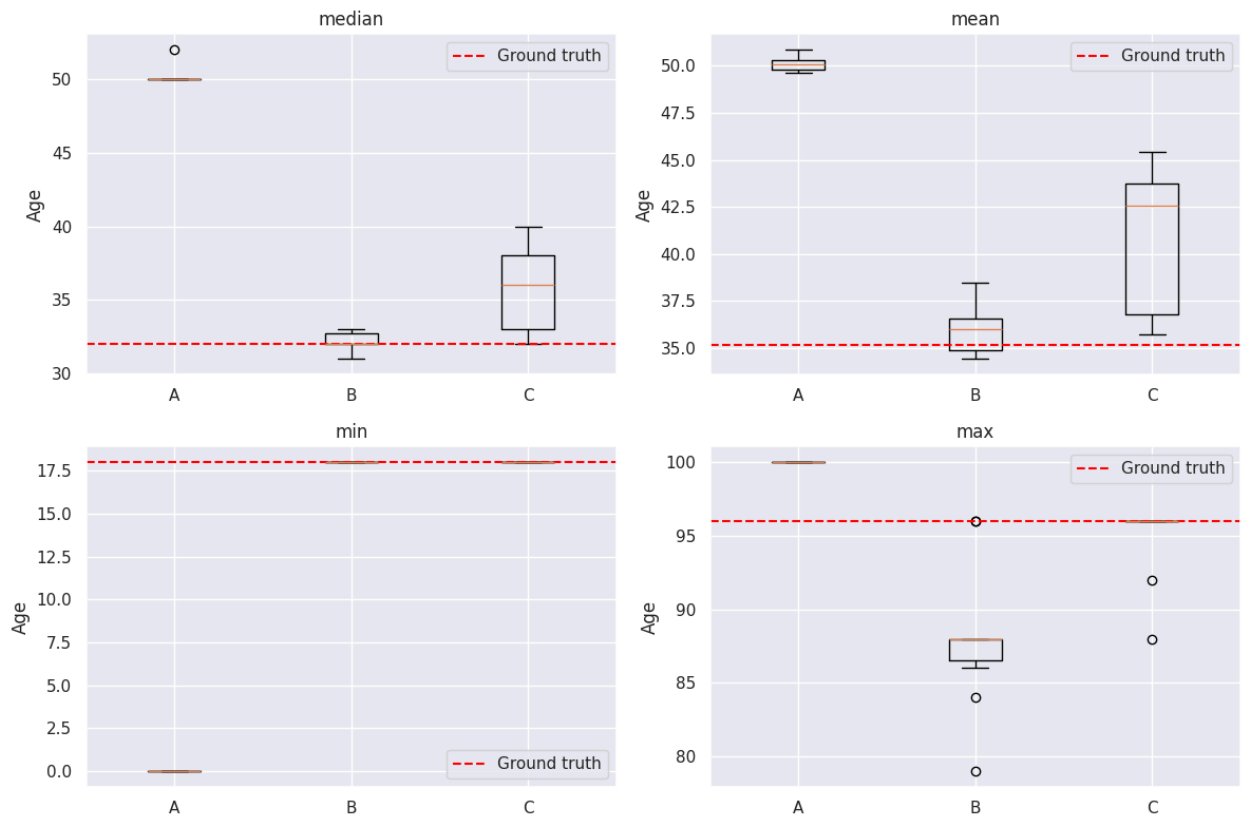
Mutual Information for df_fake_D

In the real dataset, there are clear dependencies between variables, particularly between child and parent attributes, as indicated by higher mutual information values.

For method C (k=1), the overall mutual information values are noticeably smaller, and many dependencies are weakened (for example, parent 1 with child 1, and parent 2 with child 2). This suggests that the model with k=1 is not expressive enough to capture complex relationships between attributes. Additionally, the differential privacy noise further reduces the strength of these dependencies.

For method D, still some dependencies are weakened compared to the real dataset, but different ones. Increasing the model complexity (larger k) changed the preserved attribute relationships, but there is not a clear improvement.

(b)

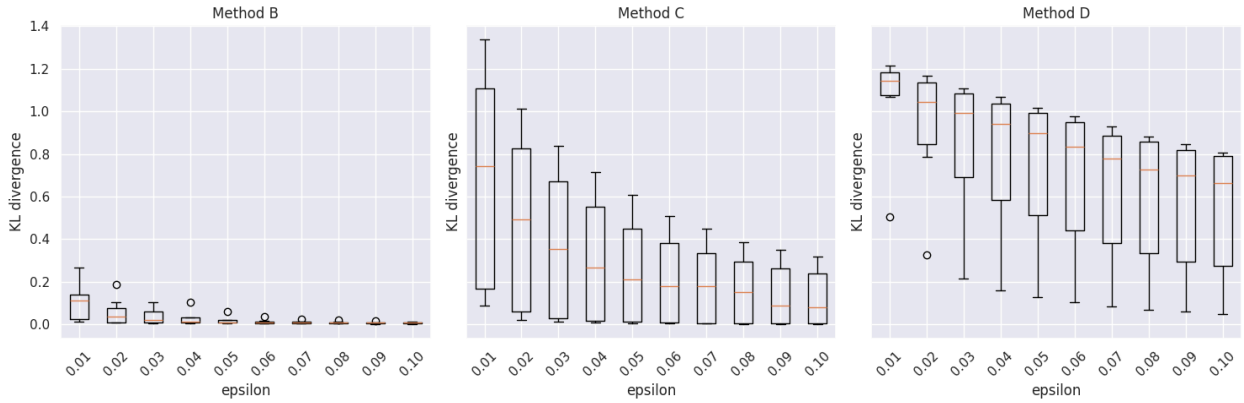


Method A shows very low variability across all statistics, as the boxplots are extremely narrow. However, the values are consistently far from the ground truth.

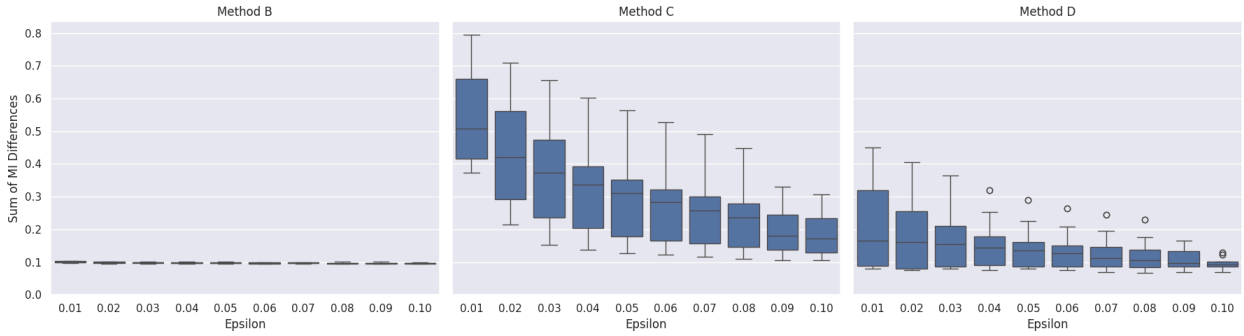
Method B shows both low variability and high accuracy. The boxplots are tightly concentrated and closely aligned with the ground truth values. The results are stable across different random seeds.

Method C shows higher variability, especially for the mean and median. It is not as accurate as method B.

(c)



KL divergence (race)



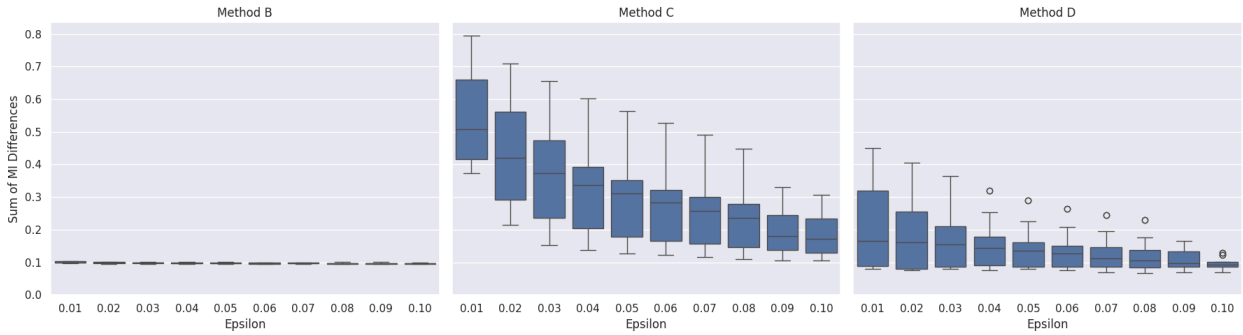
MI difference

Above is the result for the real dataset. Both KL divergence and mutual information difference decrease as ε increases for all methods. This indicates that a larger ε (weaker privacy) leads to synthetic data that more closely resembles the real dataset.

For Method B (independent attribute model), the KL divergence is consistently very low, suggesting that it approximates the marginal distribution of the race attribute well. However, the mutual information difference remains nearly constant, indicating that this method fails to capture dependencies between attributes.

Method C generally achieves lower KL divergence, while Method D yields smaller mutual information differences. This suggests that Method C better captures marginal distributions, whereas Method D is more effective at preserving correlations between attributes due to its higher model complexity.

What's more, variability is larger for small ε values and decreases as ε increases, reflecting the stronger randomness introduced by higher levels of privacy noise.



Above is the result of MI difference generated from the fake dataset. Again, for Method B, the MI difference remains nearly constant across all values of ε. For methods C and D, the mutual information difference decreases as ε increases. This indicates that as the privacy constraint is relaxed (larger ε), the synthetic data better preserves the dependency structure of the original dataset. Also, the variability across runs is larger for small ε and decreases as ε increases.

Problem 4

a. Yes, the mechanism is differentially private. Suppose Y=1 means that the true answer is "yes", Y=0 means that the true answer is "no". If Y=1, the person answers "yes" no matter the result of the coin flip. The probability of answering "yes" is 1. If Y=0, the probability of answering "yes" is 0.5.
   Therefore, $\frac{P(yes|Y=1)}{P(yes|Y=0)} = \frac{1}{0.5} = 2$. So e^{ϵ} = 2. $\epsilon = ln\,2$. The mechanism satisfies ln2 differential privacy.
b. Suppose Y = 1 means that the true answer is "yes", and Y = 0 means that the true answer is "no". The person has a probability of 0.5 to tell the truth (when D1 comes up with a value smaller than 4). Otherwise, the person rolls D2. Then, the person has ⅓ * ½ = ⅙ probability to respond "yes", and ⅔ * ½ = ⅓ probability to respond "no".
   So, if the true answer is "yes", P(yes|Y=1) = ½ * 1 + ⅙ = ⅔. P(no|Y=1) = ½ * 0 + ⅓ = ⅓. If the true answer is "no", P(yes|Y=0) = ½ * 0 + ⅙ = ⅙. P(no|Y=0) = ½ * 1 + ⅓ = ⅚.
   $\frac{P(yes|Y=1)}{P(yes|Y=0)} = \frac{2/3}{1/6} = 4$. $\frac{P(no|Y=1)}{P(no|Y=0)} = \frac{1/3}{5/6} = 2/5$. 4 is larger, so e^{ϵ} = 4. $\epsilon = ln\,4$. The mechanism satisfies ln4 differential privacy.